

Solubility and Distribution Phenomena

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Chapter objectives

- ❑ After completion of this lecture, the student should be able to:
 - ❑ define solubility, saturated & unsaturated solutions and polar & non polar solvents.
 - ❑ discuss the various types of pharmaceutical solutions.
 - ❑ describe complete & partial miscibility.
 - ❑ identify the different factors controlling the solubility of pharmaceutical solutions.
 - ❑ calculate partition coefficient & understand its importance in pharmaceutical systems.

Importance of studying the phenomenon of solubility

- ❑ Understanding the phenomenon of solubility helps the pharmacist to:
 - ❑ Select the best solvent for a drug or a mixture of drugs.
 - ❑ Overcome problems arising during preparation of pharmaceutical solutions.
 - ❑ Have information about the structure and intermolecular forces of the drug.

Definitions:

❑ **True solution:** is a mixture of two or more components that form a homogenous molecular dispersion. The components are referred to the **solute** & the **solvent** .

❑ **Solute:** is the dissolved agent (less abundant part of the solution).

❑ **Solvent:** is the component in which the solute is dissolved (more abundant part of the solution).

Definitions..

- ❑ **A saturated solution:** is one in which an equilibrium is established between dissolved and undissolved solute **at a definite temperature.**
- ❑ **An unsaturated solution or subsaturated solution :** is one containing the dissolved solute in a conc. below that necessary for complete saturation at a definite temperature.

Definitions..

- ❑ **A supersaturated solution:** contains more of the dissolved solute than it would normally contain in a saturated state at a definite temperature.
- ❑ Some salts (e.g. sodium thiosulfate) can be dissolved in large amounts at an elevated temperature and, upon cooling fail to crystallize from the solution; these super saturated solutions can be converted to saturated ones by seeding the solution with a crystal of solute.

Definitions...

☐ Solubility:

☐ **in a *quantitative* way:** it is the concentration of solute in a saturated solution *at a certain temperature*.

☐ **in a *qualitative* way:** it is the spontaneous interaction of two or more substances (solute & solvent) to form a homogeneous molecular dispersion.

Solubility expressions

a) The solubility of a drug can be expressed in terms of:

- ☐ Molarity

- ☐ Normality

- ☐ Molality

- ☐ Mole fraction

- ☐ Percentage (% w/w, % w/v, % v/v).

Solubility expressions

b) The USP lists the solubility of drugs as the **number of ml of solvent** in which 1 g of solute will dissolve.

E.g. 1g of boric acid dissolves in 18 mL of water, and in 4 mL of glycerin.

c) Substances whose solubility values are not known are described by the following terms:

Terms of solubility

Term	Parts of solvent required for 1 part of solute
Very soluble	Less than 1 part
Freely soluble	1 to 10 parts
Soluble	10 to 30 parts
Sparingly soluble	30 to 100 parts
Slightly soluble	100 to 1000 parts
Very slightly soluble	1000 to 10 000 parts
Practically insoluble	More than 10 000 parts

Solvent Solute Interaction

- ❑ The pharmacist knows that water is a good solvent for salts, sugars, and similar compounds.
- ❑ Mineral oil and benzene are often solvents for substances that are normally only slightly soluble in water.
- ❑ There empirical findings are summarized in the statement: "**like dissolves like**".

Polar solvent

- ❑ The solubility of a drug is due to the polarity of the solvent, i.e., to its dipole moment.
- ❑ Polar solvents dissolve ionic solutes and other polar substances.
- ❑ Accordingly, water mixes in all proportions with alcohol and dissolves sugars and other polyhydroxy compounds.

Polar solvent...

- ❑ Water dissolves phenols, alcohols, aldehydes, ketones, amines, and other oxygen and nitrogen-containing compounds that can form hydrogen bonds with water.
- ❑ When ethyl alcohol and water are mixed, the hydrogen bonds between the water molecules are replaced partly by hydrogen bonds between water and alcohol molecules.

Polar solvent...

- ❑ the solubility of a substance also depends on the ratio of the polar to non-polar groups of the molecule.
- ❑ As the length of a non-polar chain of an aliphatic alcohol increases, the solubility of the compound in water decreases.
- ❑ Polyhydroxy compounds as glycerin, tartaric acid, PEG are water soluble.

Non polar Solvents

- ❑ ionic and polar solutes are not soluble or are only slightly soluble in non polar solvents.
- ❑ Non polar compounds can dissolve non polar solutes
- ❑ Oils and fats dissolve in carbon tetrachloride, benzene, and mineral oil.
- ❑ Alkaloidal bases and fatty acids also dissolve in non polar solvents.

Semipolar Solvents

- ❑ Semipolar solvents, such as ketones and alcohols, can induce a certain degree of polarity in non polar solvent molecules.
- ❑ Semipolar compounds may act as intermediate solvents to bring about miscibility of polar and non polar liquids.

Semipolar Solvents....

- ❑ Propylene glycol increase the mutual solubility of
 - ❑ water and peppermint oil and
 - ❑ water and benzyl benzoate.
- ❑ Benzene which is readily polarizable becomes soluble in alcohol
- ❑ Acetone increases the solubility of ether in water.

Intermolecular force in solution formation

❑ Intermolecular forces are responsible for the **physical state** of a compound (solid, liquid or gas).

❑ Van der Waals Forces

✓ Dipole interactions (keesom forces)

✓ Dipole-Induced Dipole Forces (Debye forces)

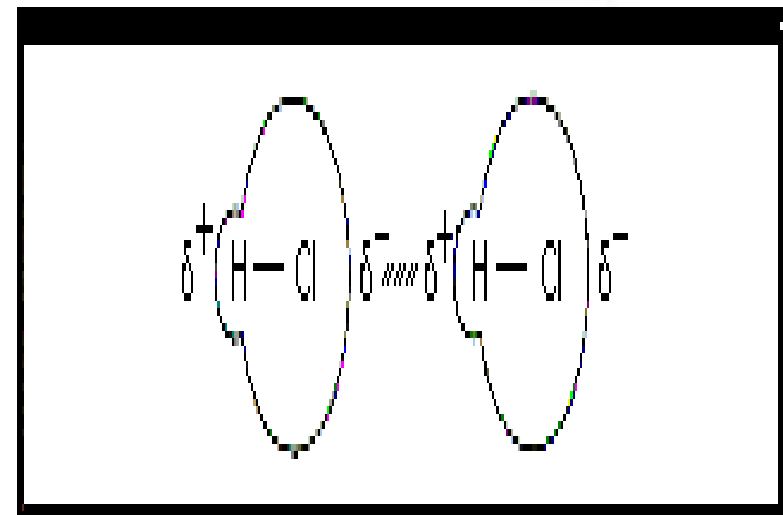
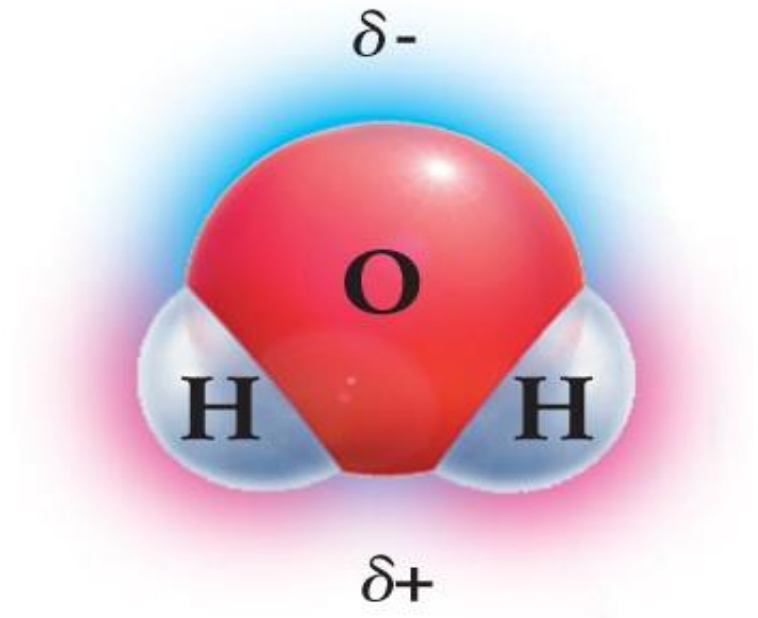
✓ Dispersion Forces (london force)

❑ Ion-Dipole Forces

❑ Hydrogen Bonds

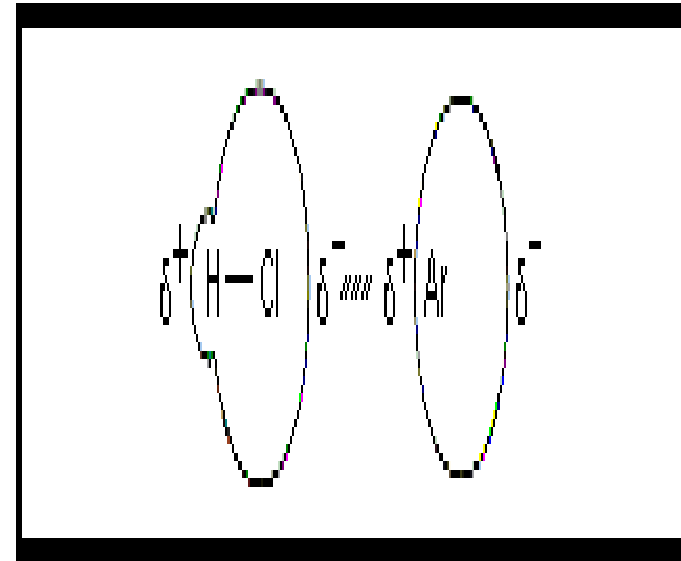
Dipole-Dipole Forces

- ❑ A polar molecule that has two poles
- ❑ These molecules attracted to each other.
- ❑ The positive end of one is attracted to the negative end of the other
- ❑ important when the molecules are close to each other.



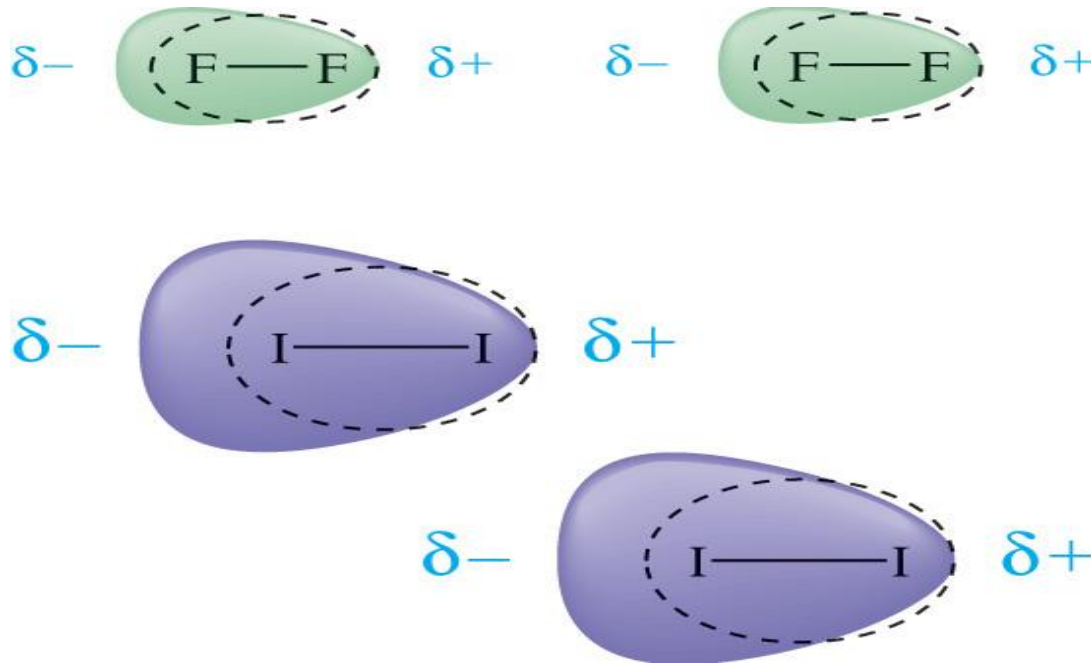
Dipole-Induced Dipole Forces

- ❑ A permanent dipole polarize charge in neighboring molecule
- ❑ The electrons on an argon atom are distributed homogeneously around the nucleus of the atom.
- ❑ When an argon atom comes close to a polar HCl molecule, the electrons can shift to one side of the nucleus to produce a very small dipole moment that lasts for moments



Dispersion Forces

- ❑ Caused by the motion of electrons
- ❑ Increase as the number of electrons increases.
- ❑ Weakest of all intermolecular forces



Ion-Dipole Forces

Ion-Dipole Interaction

Na^+



I^-



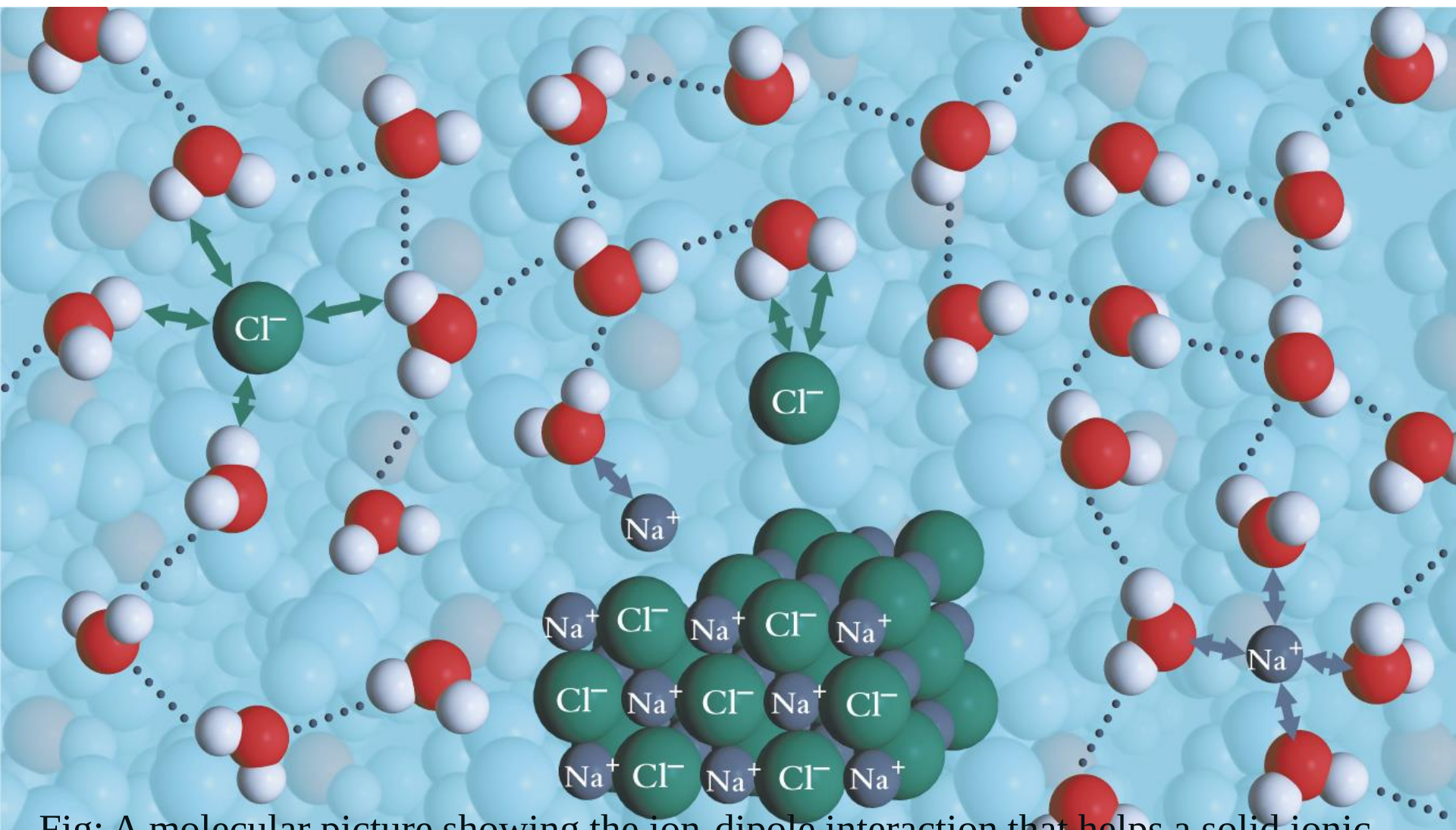


Fig: A molecular picture showing the ion-dipole interaction that helps a solid ionic crystal dissolve in water. The arrows indicate ion-dipole interactions

Hydrogen bonding

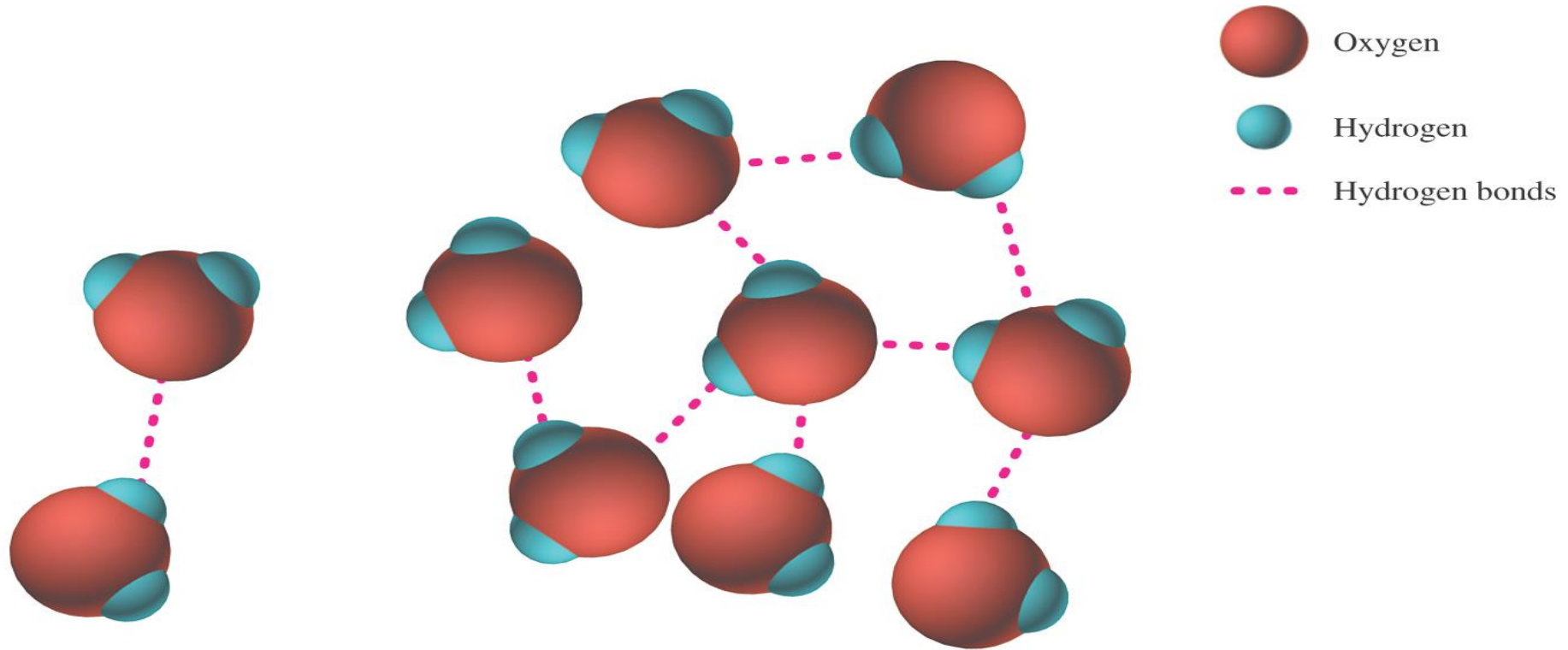
- ❑ A hydrogen bond is formed between polar molecules that contain hydrogen atom covalently bonded to a small, highly electronegative atom: F, O, N.
- ❑ Hydrogen can move in close to an electronegative atom and form an electrostatic type of union
 - ❑ b/c of its small size and its large electrostatic field



Hydrogen bonding...

- the strongest of all intermolecular forces
- Hydrogen bonds are responsible for the physical properties of many biological substances (e.g. water)

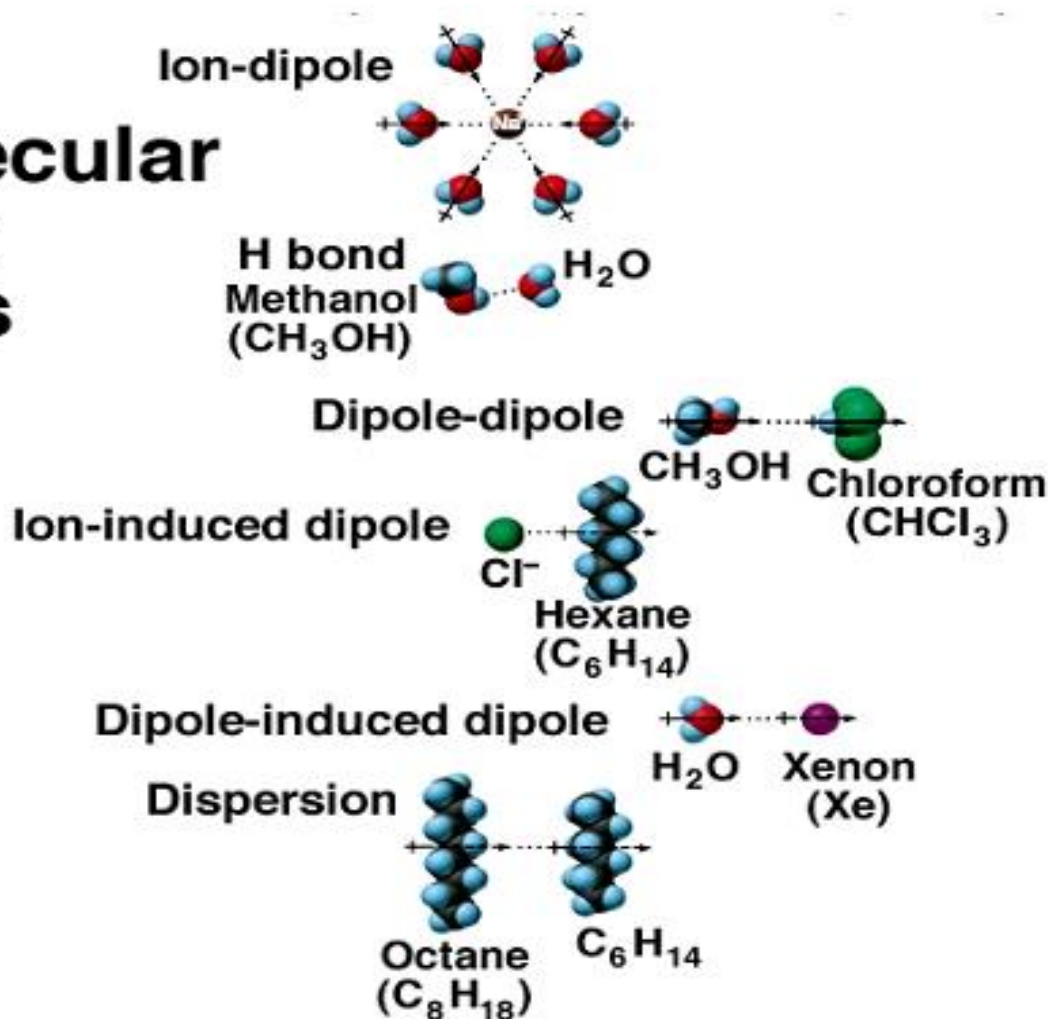
Hydrogen bonding...



Water in the liquid and solid states exists as groups in which the water molecules are linked together by hydrogen bonds

What happens when solutes dissolve

Intermolecular Forces in Solutions



Assignment : What type(s) of intermolecular forces exist between each of the following molecules?

1. HBr
2. SO₂
3. H₂O
4. CO₂
5. CH₄

Solution formation

- ❑ The main factors affecting the dissolution of a solute in a solvent are
 - ❑ solute-solute interactions
 - ❑ solvent-solvent interactions
 - ❑ solvent-solute interactions
- ❑ Dissolution will be favored when the first two are relatively small and the third is relatively large

Three Steps in making a Solution

Step #1 :

- Breaking up the solute into individual components:(expanding the Solute)

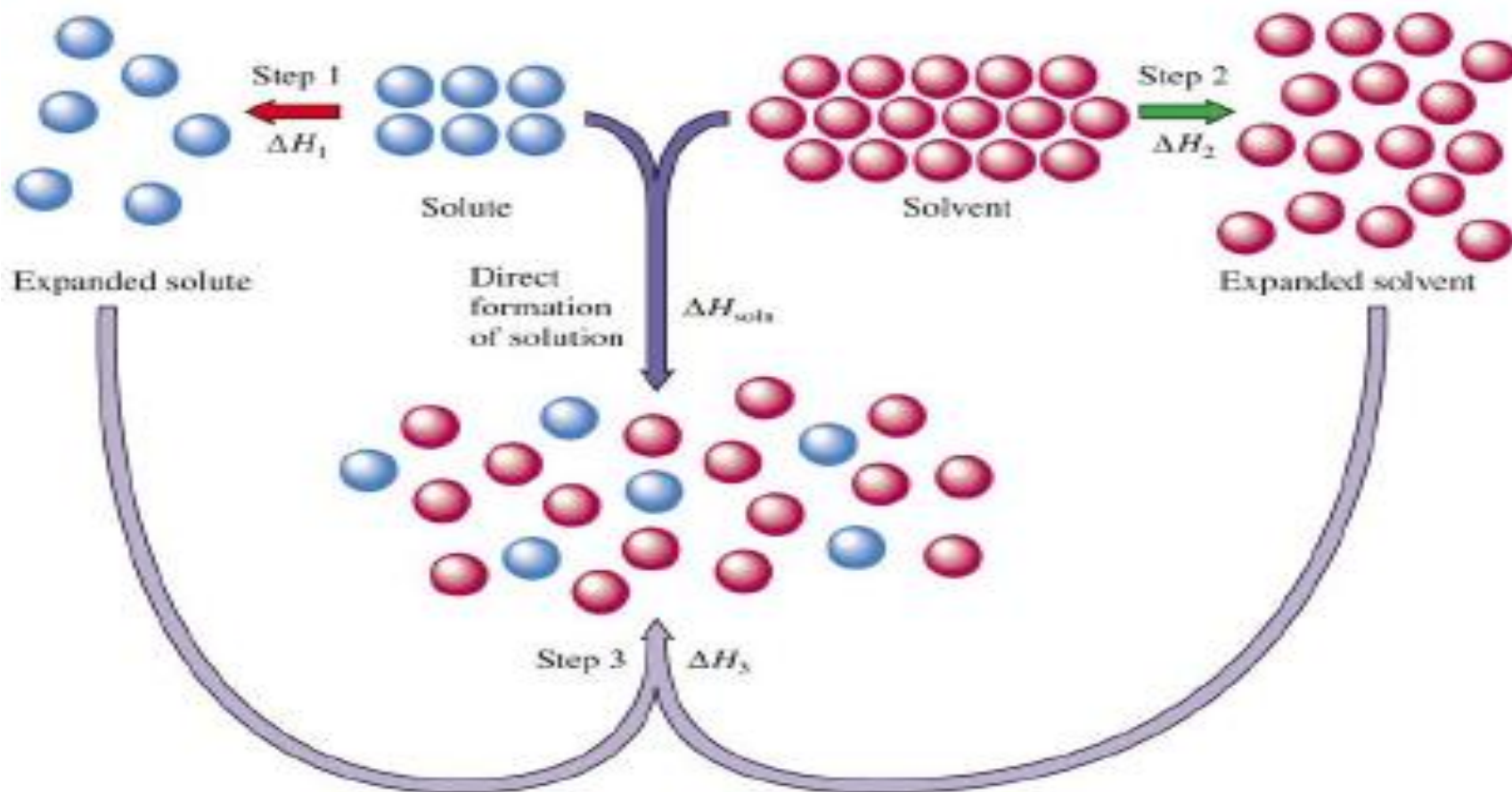
Step #2 :

- Overcoming intermolecular forces in the solvent to make room for the solute: (expanding the solvent)

Step #3 :

- Allowing the solvent and solute to interact and form the solution.

Three Steps in making a Solution...



Solution formation...

Enthalpy of Solution (heat of solution)

❑ Consider solution formation as three hypothetical steps, each with an enthalpy change associated with it.

Step 1: Separating solute molecules ΔH^1

Step 2: Separating solvent molecules ΔH^2

Step 3: Allowing the solute and solvent to interact to form the solution ΔH^3

$$\Delta H^{\text{solution}} = \Delta H^1 + \Delta H^2 + \Delta H^3$$

Solution formation...

A. Ideal solutions ($\Delta H^{\text{solution}} = 0$, $\Delta V^{\text{solution}} = 0$)

- ❑ All intermolecular forces are of comparable strength
- ❑ No net energy change on solution formation
- ❑ Volume of solution is the sum of the volumes of the solute and solvent
- ❑ Some liquids come close to forming ideal solutions. Ex:
toluene in benzene

Solution formation...

B. $\Delta H^3 > \Delta H^1 + \Delta H^2$ ($\Delta H^{\text{solution}} < 0$, $\Delta V^{\text{solution}} < 0$)

- ☐ intermolecular forces between solute and solvent particles are greater than other intermolecular forces
- ☐ Solution formation is exothermic
- ☐ Volume of the solution is less than the sum of the solute and solvent volumes.
- ☐ Non ideal

Solution formation...

C. $\Delta H^3 < \Delta H^1 + \Delta H^2$ ($\Delta H^{\text{solution}} > 0$, $\Delta V^{\text{solution}} > 0$)

- ☐ Intermolecular forces between solute and solvent particles are weaker than other intermolecular forces.
- ☐ Solution formation is **endothermic**.
- ☐ Volume of the solution is greater than the sum of the solute and solvent volumes
- ☐ Non ideal

Solution formation...

D. $\Delta H_3 < < < \Delta H_1 + \Delta H_2$

- ❑ The intermolecular forces between solute and solvent are so much smaller than the sum of the other intermolecular forces
- ❑ the two remain as separate (**heterogeneous**) phases.

Methods of expressing the concentration of a solution

- ❑ *Concentration* is a general term referring to the quantity of solute used to prepare a given amount of solution

1. “Parts-per” concentration

- ❑ based on the quantity of solute in a fixed quantity of solution.
- ❑ The “quantities” can be expressed in **weight**, in **volume**, or both (the weight of solute in a given volume of solution.)
- ❑ the abbreviations (w/w), (v/v) and (w/v) are used.

Methods of expressing the concentration of a solution

Assignment1:

1. Find the percent (w/w) concentration of a solution containing 4.5 g of sucrose in 90 mL of water.
2. Describe how you would prepare 30 g of a 20 percent (w/w) solution of KCl in water.

Methods of expressing the concentration of a solution

2. Molarity: mole/volume basis

- ❑ Molar concentration (molarity) is the number of moles of solute per liter of *solution*.

Assignment 2:

- ❑ Calculate the molarity of a 60 % (w/w) solution of ethanol ($\text{C}_2\text{H}_5\text{OH}$) in water whose density is 0.8937 g mL^{-1}

Methods of expressing the concentration of a solution

3 Mole fraction: mole/mole basis

- The mole fraction of substance *i* in a mixture is defined as

$$\chi_i = \frac{n_i}{\sum_j n_j}$$
 in which n_j is the number of moles of substance *i*, and the summation is over all substances in the solution.

- Mole fractions run from zero (substance not present) to unity (the pure substance).

Assignment 3:

What fraction of the molecules in a 60 % (w/w) solution of ethanol in water consist of H₂O?

Methods of expressing the concentration of a solution

4. Molality: mole/weight basis

- The molality of a solution is defined as the number of moles of solute divided by the mass of the solvent
- The [SI-unit](#) for molality is mol/kg.
- it offers a more precise description of concentration because it is unaffected by temperature.

Types of solutions

❑ Solutions of pharmaceutical importance include:

❑ gases in liquids

❑ liquids in liquids

❑ solids in liquids

Solubility of gases in liquids

- ❑ The solubility of gas in a liquid represents the concentration of dissolved gas in the liquid when it is in equilibrium with some of the pure gas above the solution
- ❑ The amount of gas that will dissolve in a liquid is determined by
 - the natures of the two components
 - temperature and
 - pressure

Solubility of gases in liquids...

Factors affecting the solubility of gases in liquids

Effect of pressure

- ❑ Provided that no reaction occurs between the gas and liquid then the effect of pressure is indicated by Henry's law,
- ❑ “states that at constant temperature the solubility of a gas in a liquid is directly proportional to the pressure of the gas above the liquid”

Solubility of gases in liquids...

- ❑ The law may be expressed as

$$w = kp$$

- ❑ where w is the mass of gas dissolved by unit volume of solvent at an equilibrium pressure p and
- ❑ P is the partial pressure in mmHg of the undissolved gas above the solution and can be obtained by subtracting the vapour pressure of the solvent from the total pressure of the solution
- ❑ k is a proportionality constant (the solubility coefficient)

Solubility of gases in liquids...

- ❑ Henry's law is most applicable at high temperatures and low pressures
 - ❑ but when solubility is low it provides a satisfactory description of the behavior of most systems at normal temperatures and reasonable pressures
- ❑ Henry's eqn. also applies to the solubility of each gas in a solution of several gases in the same liquid

Solubility of gases in liquids...

Effect of Temperature

- ❑ The solubility of most gases in liquids decreases as the temperature rises
 - ❑ Due to the greater tendency of the gases to expand in comparison to the solvent
- ❑ This provides a means of removing dissolved gases
 - ❑ E.g. water for injections free from either carbon dioxide or air

Solubility of gases in liquids...

Effect of electrolyte and non-electrolyte

- ❑ Solubility of gases in a gases liquids is generally reduced by the addition of electrolyte (NaCl) and non-electrolyte (sugar)
 - ❑ by a 'salting out' process
- ❑ This is due stronger affinity b/n solvent and electrolyte or non-electrolyte than b/n the solvent and the gas

Solubility of gases in liquids...

Effect of chemical reaction

- ❑ Solubility of gas in liquids increases if there is chemical rxn
b/n a gas and the solvent
- ❑ E.g. HCl gas reacts with water by hydrogen bonding when it
dissolves in water.

Solubility of liquids in liquids

Ideal solution

- ❑ Are those which obeys Rault's law over the whole ranges of composition at all T
- ❑ Rault's law expressed as

$$P_A = P_A^0 X_A$$

Where P_A is the partial pressure of a component in a liquid mixture

P_A^0 is the vapour pressure in pure state

X_A is mole fraction of the component in the solution

Solubility of liquids in liquids...

- ❑ Mixtures of nearly adjacent cpds. in homologous series at low T are ideal solution
 - ❑ E.g benzene-toluene, methanol-ethanol, hexane-heptane

Real solution

- ❑ Are those which show deviation from Raoult's law Deviation may be
 - ❑ Negative deviation
 - ❑ Positive deviation

Solubility of liquids in liquids...

- ❑ Two liquids are either completely or partially miscible
 - ❑ Completely miscible liquids mix in all proportions
 - ❑ Partially miscible liquids form two immiscible layers, each of which saturated solution of one liquid in other
 - ❑ The two liquids phases are conjugate liquid phases
- ❑ The mutual solubility of partially miscible liquids influenced by T

Solubility of liquids in liquids...

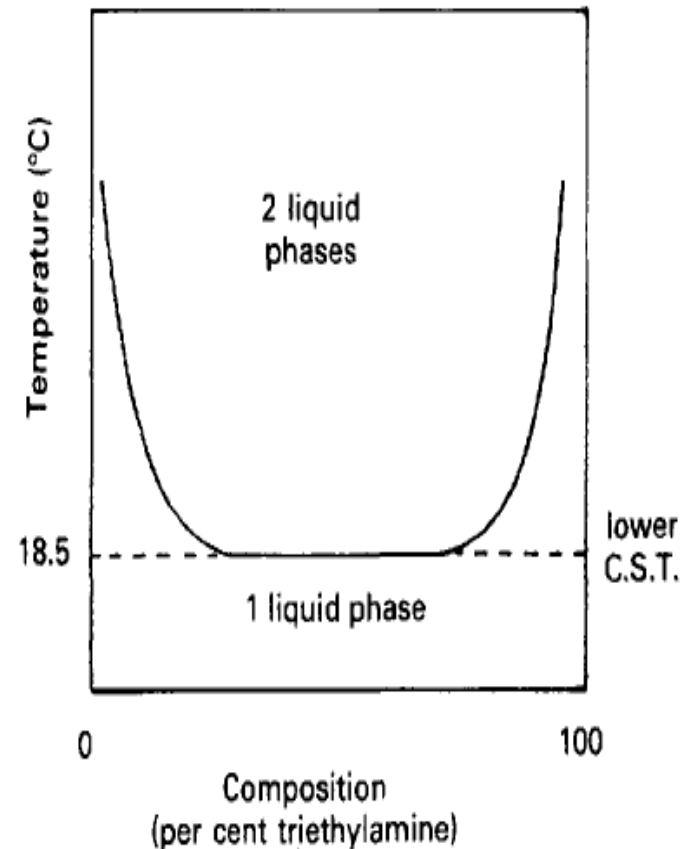
Systems showing an increase in miscibility with rise in temperature

- ❑ A positive deviation from Raoult's law arises from
 - ❑ a difference in the cohesive forces
- ❑ This difference becomes more marked as the temperature decreases, and result in
 - ❑ a decrease in miscibility sufficient to cause the separation of the mixture into two phases.
- ❑ At CST homogenous phase results for any composition
- ❑ E.g. phenol and water

Solubility of liquids in liquids...

Systems showing a decrease in miscibility with rise in temperature

- ❑ A few mixtures exhibit a LCST e.g. triethylamine and water, paraldehyde and water.
- ❑ Showed negative deviation from Raoult's law, and miscibility therefore increases as the temperature falls

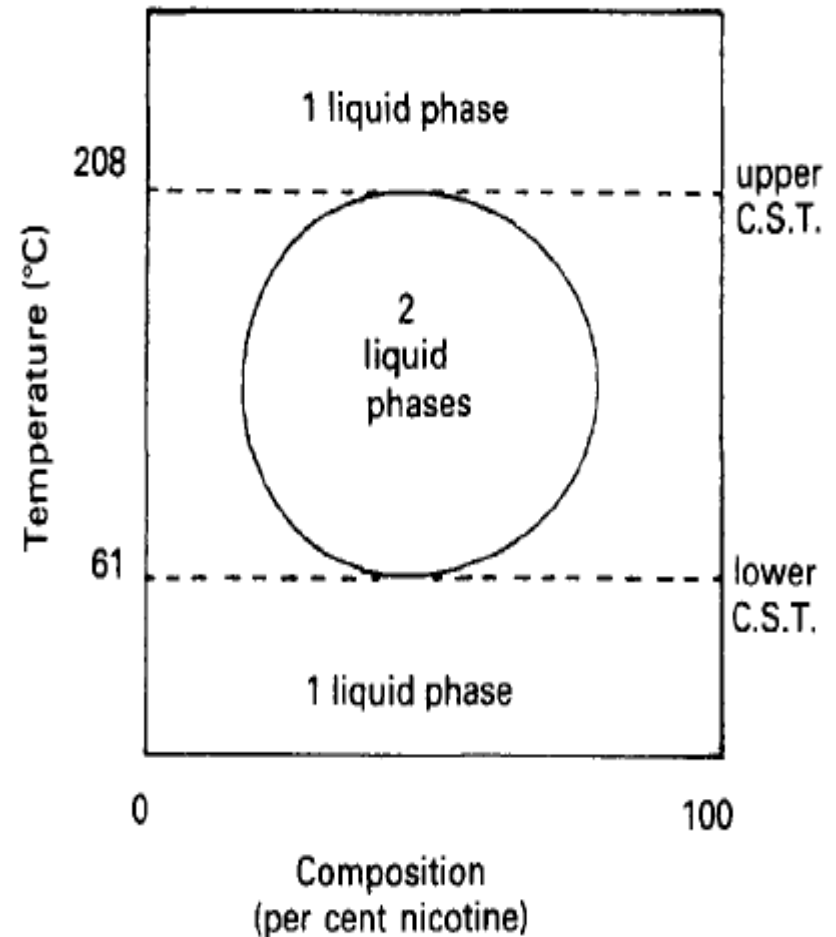


Solubility of liquids in liquids...

Systems showing upper and lower critical solution temperatures

❑ e.g. nicotine and water

❑ UCST=208 and LCST=61



Solubility of liquids in liquids...

Some types of mix have neither and UCST nor LCST

- ❑ These mix remain partially miscible over the entire T range
- ❑ E.g. ethyl ether and water

solubility of solids in liquids

Ideal solution

- ☐ No net energy change on solution formation
- ☐ All intermolecular forces are of comparable strength
- ☐ Volume of solution is the sum of the volumes of the solute and solvent

solubility of solids in liquids

Non-ideal solution

- ❑ There is net energy change on solution formation
- ❑ This non-ideality is due to various attractive forces involved
b/n solute, solvent and solute-solvent molecules
- ❑ Volume of the solution is $<$ or $>$ the sum of the solute and solvent volumes.

Factors affecting the solubility of solids in liquids

Temperature

- ❑ If ΔH is positive the dissolution process is usually an endothermic one
 - ❑ a rise in temperature will lead to an increase in the solubility
- ❑ If ΔH is negative the dissolution process is exothermic
 - ❑ an increase in temperature will give rise to a decrease in solubility

Factors affecting the solubility of solids in liquids ...

- ❑ Solubility curves are often used to describe the effect of temperature on a given system
- ❑ Most of the curves are continuous
 - ❑ however, abrupt changes in slope may be observed with some systems if a change in the nature of the dissolving solid occurs at a specific transition temperature.
- ❑ E.g $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ up to 32.5°C , its dissolution in water is an endothermic process
- ❑ Above 32.5°C the solid is converted into the anhydrous form, Na_2SO_4 , and the dissolution of this compound is an exothermic process.

Factors affecting the solubility of solids in liquids ...

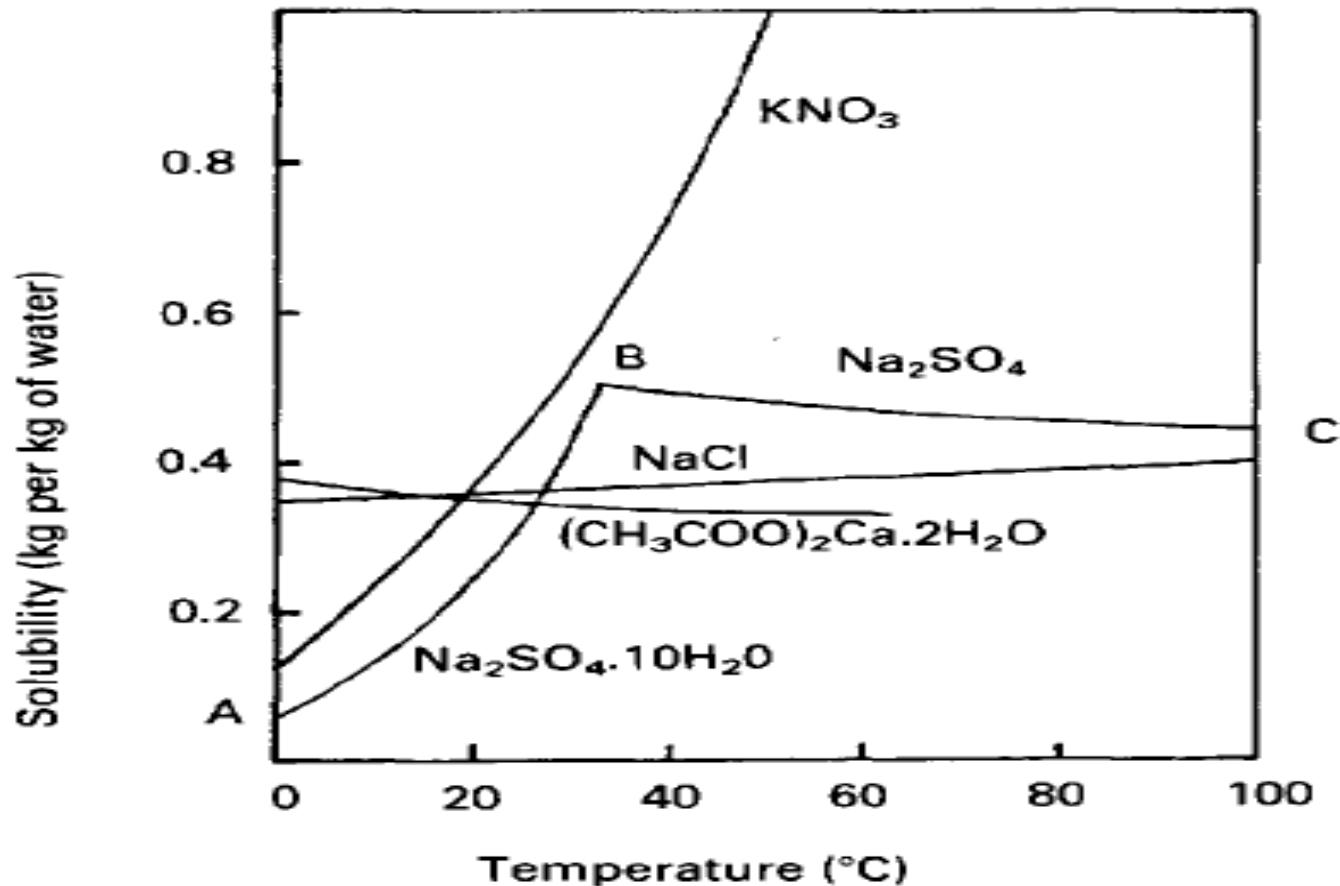


Fig. Solubility curves for various substances in water

Factors affecting the solubility of solids in liquids ...

Molecular structure of solute

- ❑ a small change in the molecular structure of a compound can have a marked effect on its solubility in a given liquid
- ❑ the introduction of hydroxyl group can produce a large improvement in water solubility
- ❑ E.g. more than 100-fold difference in the solubility's of phenol and benzene

Factors affecting the solubility of solids in liquids ...

- ❑ The conversion of a weak acid to its sodium salt leads to a much greater degree of ionic dissociation of the compound when it dissolves in water
- ❑ The overall interaction between solute and solvent is markedly increased and the solubility consequently rises
- ❑ E.g. the aqueous solubility's of salicylic acid and its sodium salt, which are 1:550 and 1:1, respectively

Factors affecting the solubility of solids in liquids ...

- ❑ The reduction in aqueous solubility of a parent drug by its esterification
- ❑ Such a reduction in solubility may provide a suitable method for:
 - ❑ masking the taste of a parent drug, e.g. chloramphenicol palmitate is used in paediatric suspensions
 - ❑ protecting the parent drug from excessive degradation in the gut, e.g. erythromycin propionate

Factors affecting the solubility of solids in liquids ...

Nature of solvent: cosolvents

- ❑ 'like dissolves like'
- ❑ mixtures of solvents are often used in pharmaceutical practice to obtain aqueous-based systems that contain solutes in excess of their solubility's in pure water
 - ❑ This is achieved by using cosolvent (e.g. ethanol or propylene glycol) which are miscible with water and better solvents for the solute
 - ❑ Work by decreasing the interfacial tension or by altering the dielectric constant of the medium

Factors affecting the solubility of solids in liquids ...

Crystal characteristics: polymorphism and solvation

- ❑ Many drugs exhibit polymorphism, e.g. steroids, barbiturates and sulphonamides
- ❑ The effect of polymorphism on solubility is particularly important
 - ❑ it provides a means of increasing the solubility of a crystalline material and hence its rate of dissolution by using a metastable polymorph
- ❑ Conversion to the less soluble and most stable polymorph may contribute to the growth of crystals in suspension formulations.

Factors affecting the solubility of solids in liquids ...

- ❑ Amorphous form lead to an increase in the solubility of a drug
- ❑ the lattice structures of crystalline materials may be altered by the incorporation of molecules of the solvent from which crystallization occurred.
 - ❑ The resultant solids are called solvates
 - ❑ the phenomenon is referred as solvation
 - ❑ the solubility's of solvated and unsolvated crystals are different

Factors affecting the solubility of solids in liquids ...

- ❑ hydrated crystals tend to exhibit a lower aqueous solubility than their unhydrated forms
 - ❑ if a **hydrate** is formed, then the interaction between the substance and water that occurs in the crystal phase reduces the amount of energy liberated when the solid hydrate dissolves in water
- ❑ The aqueous solubilities of non-aqueous solvates are often greater than those of the unsolvated forms

Factors affecting the solubility of solids in liquids ...

Particle size of the solid

❑ The effect of particle size is expressed as

$$\log \frac{S}{S_0} = \frac{2\gamma M}{2.303RT\rho r}$$

where S is the solubility of small particles of radius r ,

❑ S_0 is the normal solubility

❑ γ is the interfacial energy,

❑ M is the molecular weight of the solid,

❑ ρ is the density of the bulk solid,

❑ R is the gas constant and

❑ T is the temperature

Factors affecting the solubility of solids in liquids ...

- ❑ The increase in solubility with decrease in particle size ceases when the particles have a very small radius, and any further decrease in size causes a decrease in solubility.
- ❑ presence of an electrical charge on the particles and that the effect of this charge becomes more important as the size of the particles decreases.

Factors affecting the solubility of solids in liquids ...

pH

- ❑ This relationship between pH and the solubility of ionized solutes is extremely important with respect to the ionization of weakly acidic and basic drugs as they pass through GIT and experience pH changes.
- ❑ pH will affect the degree of ionization of the drug molecules, which in turn influences their solubility and their ability to be absorbed.

Factors affecting the solubility of solids in liquids ...

- ❑ The relationship between pH and the solubility and pKa value of an acidic drug is given as

$$\text{pH} = \text{pK}_a + \log \frac{(S - S_u)}{S_u}$$

- where S is the overall solubility of the drug and
- S_u is the solubility of its unionized form, i.e. $S = S_u + \text{solubility of ionized form } (S_i)$.

- ❑ For basic drugs the corresponding relationship is given as

$$\text{pH} = \text{pK}_a + \log \frac{S_u}{S - S_u}$$

Assignment

1. What is the minimum pH required for the complete solubility of the drug in a solution containing 6 g of phenobarbitone sodium (mwt 254) in 100 ml of a 30%v/v alcoholic solution (given pK_a of phenobarbitone in 30%v/v alcohol = 7.92 and S_U of phenobarbitone in 30%v/v alcohol = 0.0276 M).
2. Below what pH will free phenobarbitone begin to precipitate from a solution having 1 g of phenobarbitone sodium (mwt 254) per 100 ml at 25 °C? The molar solubility of phenobarbitone is 0.0050 and the pK_a = 7.41 at 25 °C.

Factors affecting the solubility of solids in liquids ...

Common ion effect

- ❑ The equilibrium in a saturated solution of a sparingly soluble salt in contact with undissolved solid may be represented as



- ❑ From the Law of Mass Action the equilibrium constant K for this reversible reaction is given by

$$K = \frac{[\text{A}^+][\text{B}^-]}{[\text{AB}]}$$

Factors affecting the solubility of solids in liquids ...

- ❑ As the concentration of a solid may be regarded as being constant, then

$$K'_s = [A^+][B^-]$$

- ❑ where K'_s is a constant which is known as the solubility product of compound AB
- ❑ These equations for the solubility product are only applicable to solutions of sparingly soluble salts

Factors affecting the solubility of solids in liquids ...

- ❑ If K'_s is exceeded by $[A^+][B^-]$, then the equilibrium moves towards the left in order to restore the equilibrium, and solid AB is precipitated.
- ❑ $[A^+][B^-]$ will be increased by the addition of more A^+ ions produced by the dissociation of another compound
 - ❑ Solid AB will be precipitated and the solubility of this compound is therefore decreased.

Factors affecting the solubility of solids in liquids ...

Effect of different electrolytes on the solubility of sparingly soluble electrolyte

- ❑ The solubility of a sparingly soluble electrolyte may be increased by the addition of a second electrolyte that does not possess ions common to the first, i.e. a different electrolyte.
- ❑ They lower the activity coefficient

Factors affecting the solubility of solids in liquids ...

Effect of non-electrolytes on the solubility of electrolytes

- ❑ The solubility of electrolytes depends on the dissociation of dissolved molecules into ions.
- ❑ The ease of this dissociation is affected by the dielectric constant of the solvent, which is a measure of the polar nature of the solvent.
- ❑ Liquids with a high dielectric constant (e.g. water) are able to reduce the attractive forces that operate between oppositely charged ions produced by dissociation of an electrolyte.

Factors affecting the solubility of solids in liquids ...

- If a water-soluble non-electrolyte (e.g. alcohol) is added to an aqueous solution of a sparingly soluble electrolyte, the solubility of the latter is decreased
 - b/c it lowers the dielectric constant of the solvent and ionic dissociation of the electrolyte becomes more difficult.

Factors affecting the solubility of solids in liquids ...

Effect of electrolytes on the solubility of non-electrolytes

- Non-electrolytes do not dissociate into ions in aqueous solution, and in dilute solution the dissolved species therefore consists of single molecules.
- Their solubility in water depends on the formation of weak intermolecular bonds between their molecules and those of water.
 - The presence of a very soluble electrolyte will reduce the solubility of a non-electrolyte by competing for the aqueous solvent and breaking the intermolecular bonds between the non-electrolyte and the water.

Factors affecting the solubility of solids in liquids ...

Complex formation

- The apparent solubility of a solute in a particular liquid may be increased or decreased by the addition of a third substance which forms an intermolecular complex with the solute.
- The solubility of the complex will determine the change in the solubility of the original solute

Factors affecting the solubility of solids in liquids ...

Solubilizing agents

- These agents are capable of forming large aggregates or micelles in solution when their concentrations exceed certain values
- In aqueous solution the centre of these aggregates resembles a separate organic phase and organic solutes may be taken up by the aggregates, thus producing an increase in their solubilities in water
 - This phenomenon is known as solubilization

Factors affecting the solubility of solids in liquids ...

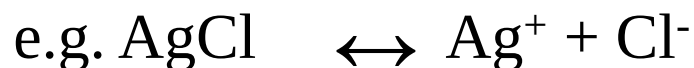
- A similar phenomenon occurs in organic solvents containing dissolved solubilizing agents, because the centre of the aggregates in these systems constitutes a more polar region than the bulk of the organic solvent.
- If polar solutes are taken up into these regions their solubilities in the organic solvents are increased

Solubility of strong electrolyte

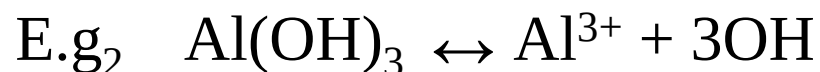
- *Strong electrolyte* dissociates completely
- A rise in temperature increases the solubility of a solid that absorbs heat...**endothermic process**
- Most substances undergo endothermic process
- NaCl does not evolve or absorb an appreciable amount of heat when dissolves in water.

Solubility of slightly soluble electrolyte

- The solubility is described by a special constant, solubility product(K_{sp})



$$K_{sp} = \frac{(\text{Ag}^+)(\text{Cl}^-)}{\text{AgCl}} = [\text{Ag}^+] [\text{Cl}^-]$$



$$K_{SP} = [\text{Al}^{3+}] [\text{OH}^-]^3$$

- Addition of common ion.....reduce solubility
- Salts having non common ion ...increase the solubility (due to complex formation)
- e.g. solubility of iron increased by Na citrate

Solubility of weak electrolyte

- Molecular compounds that produce a small concentration of ions when dissolved in H_2O
- Carboxylic acids containing more than 5 carbon atom... relatively insoluble in water...
 - ✓ react with dilute NaOH , carbonates, and bicarbonates to form soluble salts
- Fatty acids containing more than 10 carbon atoms form soluble soaps with the alkali metals.

Solubility of weak electrolyte

- Aromatic acids react with dilute alkalis to form water-soluble salts,
- Phenol is weakly acidic and only slightly soluble in water but quite soluble in dilute NaOH solution
- Organic compounds containing a basic nitrogen atom in the molecule are important in pharmacy.
 - ✓ alkaloids, antihistamines, local anesthetics...
 - ✓ soluble in dilute solutions of acids e.g. atropine sulfate and tetracaine hydrochloride

Effect of pH on Solubility of a weak electrolyte



$$S_u = [\text{HA}]$$

$$[\text{HA}]_T = \text{unionized } [\text{HA}] + \text{ionized}(\text{A}^-)$$

$$S = [\text{HA}] + [\text{A}^-]$$

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$$

$$S = S_u + K_a \frac{S_u}{[\text{H}_3\text{O}^+]} = S_u \left[1 + \frac{K_a}{[\text{H}_3\text{O}^+]} \right]$$

$$\text{Log}(S - S_u) = \text{log } k_a + \text{log } S_u - \text{log}([\text{H}_3\text{O}^+]^+)$$

$$\text{pH} = \text{P}k_a + \text{Log} \frac{(S - S_u)}{S_u}$$

- For basic drugs the corresponding relationship is given as

$$pH = pK_a + \log \frac{S_u}{S - S_u}$$

Distribution or partition coefficient

- ❑ Distribution or partition coefficient is a measure of the ability of a compound to distribute in two immiscible phases.
- ❑ The partition phenomenon is of paramount importance for the diffusion across skin and other epithelia.

Distribution solute between immiscible solvent

- If a substance which is soluble in both components of a mixture of immiscible liquids is dissolved in such a mixture, then, when equilibrium is attained at constant temperature, the solute will be distributed between the two liquids in such a way that the ratio of the activities of the substance in each liquid is a constant
- Nernst distribution law, which can be expressed by

$$\frac{a_A}{a_B} = \text{constant}$$

Distribution solute between immiscible solvent...

- where a_A and a_B are the activities of the solute in solvents A and B, respectively.
- When the solutions are dilute, or when the solute behaves ideally, the activities may be replaced by concentrations (C_A and C_B):

$$\frac{C_A}{C_B} = K$$

- where the constant K is the distribution coefficient or partition coefficient

Distribution solute between immiscible solvent...

- In the case of sparingly soluble substances K is approximately equal to the ratio of the solubilities (S_A and S_B) of the solute in each liquid

$$\frac{S_A}{S_B} = K$$

Distribution solute between immiscible solvent...

- Many pharmaceutical processes based on the distribution principles
 - ✓ absorption from the gastrointestinal tract after oral administration
 - ✓ drug distribution following entry into systemic circulation,
 - ✓ extraction and isolation of pure drugs after synthetic manufacturing or from crude plant sources,
 - ✓ formulation of a stable dosage form (emulsion, etc.)

Distribution solute between immiscible solvent...

Assignment:

- ❑ Succinic acid (0.15 g) dissolved in 100 ml of ether was shaken with 10 ml of water at 37°C. After equilibrium was achieved, the water layer contained 0.067 g of succinic acid. What is the **partition coefficient (K)** of succinic acid?



Thank You !